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Batteries and AI | Asia Pacific

Investor Feedback on Sodium-Ion Batteries

Investor discussions regarding sodium-ion batteries have centered on three key areas: 1) supply chain validation and scalability, 2) suitability for AIDC applications vs. Lithium Iron Phosphate (LFP) batteries, and 3) the new breed's competitive landscape .

Key Takeaways

- On Monday in Munich, Germany, CATL unveiled TENER Sodium ESS – the world's first field-validated sodium-ion battery energy storage system.
- CATL is targeting TENER Sodium shipments of 1GWh by end-2026, with global deliveries scheduled to commence in June 2027.
- >10GWh of material supply has been secured now; further supply chain expansion is on the way
- Superior rate capability vs. LFP makes batteries particularly well suited for AIDC storage, apart from cheaper cost and cold performance.
- Sodium-ion chemistry sets higher technical barriers to entry given extensive material science innovations, resulting in concentrated supply structure.

Sodium-ion batteries are better positioned than LFP in AIDC storage: Sodium-ion batteries appear particularly well suited to AIDC applications in view of their superior rate capability relative to conventional LFP batteries. Though energy density remains somewhat lower, sodium-ion chemistry generally exhibits faster ion transport kinetics, lower polarization under high-current operation, and better power retention at low temperatures. These characteristics enable stronger charge/discharge performance during frequent acceleration, braking and power-demand fluctuations, which are often seen in AIDCs.

Thus, sodium-ion batteries are especially valuable for AIDC use cases, which require rapid response, high peak power output, and repeated cycling. As a result, we believe they could offer a more compelling balance of performance, durability, and cost than LFP in AIDC applications, making them a natural early-adoption segment for sodium-ion commercialization.

See the following section – [Key Themes \(cont'd\)](#)

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Key Themes (cont'd)

Further progress, a new milestone: CATL officially launched the TENER Sodium ESS on Monday this week (the same day we published our [Global Insight report on sodium-ion batteries](#)), marking an important milestone in the commercialization of sodium-ion batteries. It is the world's first field-validated sodium-ion energy storage system, indicating that the technology, manufacturing capability, and supply chain have reached commercial readiness. Management targets cumulative shipments of 1GWh by end-2026, with initial deliveries in China scheduled to begin in September 2026 and global deliveries commencing in June 2027.

We view this announcement as a meaningful validation of sodium-ion technology. It moves the industry beyond laboratory demonstrations and pilot projects toward GWh-scale deployment. More importantly, CATL's willingness to commit to commercial volume targets suggests increasing confidence in supply-chain maturity and cost competitiveness. In our view, the TENER Sodium launch could represent a key inflection point for broader sodium-ion adoption in ESS.

Supply chain validation: While sodium-ion batteries benefit from abundant raw materials and significant overlap with existing lithium-ion manufacturing processes, investors are looking for evidence that the entire value chain – from cathode materials and hard carbon anodes to cell manufacturing and system integration – can achieve commercial-scale production with consistent quality and costs.

CATL's recent launch of the TENER Sodium ESS also provided an important validation of supply-chain readiness. Management disclosed that the company has already secured tens of thousands of tonnes of sodium-ion cathode and anode material supply, which is sufficient to support more than 10GWh of sodium-ion battery production, according to our calculation.

We expect CATL to continue expanding its materials ecosystem and supplier base, particularly given its stated target to build 40GWh of sodium-ion manufacturing capacity by the end of this year. In our view, the ability to secure upstream materials at scale may become a key competitive differentiator, further reinforcing the advantages of early movers with integrated R&D, supplier relationships, and manufacturing scale.

Higher barriers to entry in the new breed: We believe sodium-ion batteries carry higher technical barriers to entry than LFP, stemming from the extensive material science innovations required across cathode chemistry, hard-carbon anodes or anode-free solutions, electrolyte formulation, and cell architecture. Unlike LFP, whose supply chain has become largely mature and commoditized after a decade of industry scaling, sodium-ion remains in an earlier stage where performance differentiation depends heavily on proprietary materials and manufacturing know-how. This has resulted in a more concentrated industry structure, with CATL and a small number of leading players holding meaningful technology, capacity, and supply-chain advantages. As commercialization accelerates, we expect the sodium-ion market to remain more consolidated than the LFP market, supporting stronger pricing power for incumbents, with higher barriers to entry for newcomers.

Key technical challenges in producing high-quality sodium-ion batteries include:

Cathode material optimization: Sodium-ion cathodes suffer from greater structural instability than LFP, requiring sophisticated materials engineering, surface coating, and elemental doping to simultaneously improve energy density, cycle life and safety. Different pathways (layered oxides, Prussian Blue, polyanion systems) each present unique trade-offs.

Anode development: Unlike lithium batteries, which typically use mature graphite anodes, sodium-ion batteries rely predominantly on hard carbon. Achieving high initial Coulombic efficiency, low irreversible capacity loss, and consistent pore structure remains a major technical hurdle and manufacturing challenge.

Importantly, CATL has also developed an anode-free sodium-ion battery architecture for EV applications, which significantly improves energy density and narrows the gap versus LFP. Management has indicated that next-generation sodium-ion batteries are approaching high-end LFP-level energy density, potentially removing one of the last major obstacles to broader adoption in passenger vehicles. The achievement is technologically significant because anode-free designs require sophisticated control of sodium plating/stripping behavior, electrolyte stability, interface engineering and cycle-life management.

In our view, this further reinforces the notion that sodium-ion batteries are not merely a lower-cost alternative chemistry, but a technology platform requiring deep materials-science expertise and substantial R&D investment. Such capabilities are concentrated among a small number of industry leaders, potentially resulting in a more consolidated market structure than the commoditized LFP industry.

Electrolyte and SEI engineering: Sodium-ion batteries require highly customized electrolyte formulations to stabilize the solid-electrolyte interphase (SEI), improve cycle life and support wide-temperature operation. This remains a key area of proprietary differentiation among leading players.

Low-temperature and high-rate performance balancing. While sodium-ion batteries naturally exhibit strong low-temperature and high-rate characteristics, achieving these advantages without sacrificing cycle life, energy density and safety requires substantial cell-level optimization.

Manufacturing yield and consistency: Although sodium-ion production can leverage much of the lithium-ion manufacturing infrastructure, achieving automotive- and utility-grade consistency requires extensive process tuning. Yield management and cell consistency are especially important given the relatively immature state of the supply chain.

System integration and commercialization validation: Utility-scale deployments require validation of long-duration performance, thermal stability, safety and degradation characteristics. CATL's recent launch of the TENER Sodium ESS highlights the importance of proving readiness not only at the cell level, but across manufacturing, supply chain and system integration.

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Total	3,667		920			1632	

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INDUSTRY COVERAGE: China Energy & Chemicals

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/25/2026)
Jack Lu		
Bluestar Adiseo Co (600299.SS)		Rmb9.04
China Oilfield Services Ltd. (2883.HK)	E (03/28/2026)	HK\$6.43
China Oilfield Services Ltd. (601808.SS)	U (03/28/2026)	Rmb11.09
China Petroleum & Chemical Corp. (600028.SS)	E (08/19/2024)	Rmb4.55
China Petroleum & Chemical Corp. (0386.HK)	O (07/29/2025)	HK\$4.10
CNOOC (0883.HK)	O (03/17/2021)	HK\$21.22
Contemporary Amperex Technology Co. Ltd. (300750.SZ)	O (06/25/2025)	Rmb401.90
Contemporary Amperex Technology Co. Ltd. (3750.HK)	O (05/04/2026)	HK\$716.50
EVE Energy Co Ltd (300014.SZ)	E (05/31/2022)	Rmb66.81
Gotion High Tech Co Ltd (002074.SZ)	E (04/17/2023)	Rmb28.66
Guangzhou Tinci Materials Technology Co (002709.SZ)	E (01/08/2026)	Rmb55.20
Hengli Petrochemical Co Ltd (600346.SS)	++	Rmb18.21
Ningbo Ronbay New Energy Technology (688005.SS)	U (10/27/2025)	Rmb32.80
Ningxia Baofeng Energy Group Co., Ltd. (600989.SS)	E (04/15/2026)	Rmb21.17
PetroChina (601857.SS)	O (08/19/2024)	Rmb9.06
PetroChina (0857.HK)	O (03/17/2021)	HK\$8.73
REPT Battero Energy Co (0666.HK)	E (10/20/2025)	HK\$13.34

Rongsheng Petrochemical Co Ltd (002493.SZ)	E (07/29/2025)	Rmb11.71
Shanghai Putailai New Energy Tech Co Ltd (603659.SS)	U (10/27/2025)	Rmb28.55
Shenzhen Senior Technology Material Co (300568.SZ)	E (01/08/2026)	Rmb17.82
Yunnan Energy New Material Co Ltd (002812.SZ)	O (10/27/2025)	Rmb68.27
Kaylee Xu		
Jiangsu Cnano Technology Co Ltd (688116.SS)	U (03/10/2025)	Rmb40.10
Jiangsu Yangnong Chemical (600486.SS)	O (06/12/2026)	Rmb53.80
Shandong Sinocera Functional Material (300285.SZ)	E (03/23/2026)	Rmb98.40
Shenzhen Capchem Technology Co Ltd (300037.SZ)	E (06/07/2023)	Rmb90.29
Sunresin New Materials Co Ltd (300487.SZ)	E (10/25/2024)	Rmb60.15
Wanhua Chemical (600309.SS)	O (04/10/2026)	Rmb73.61
Zhejiang NHU Co. Ltd. (002001.SZ)	E (01/26/2026)	Rmb30.00

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.